

Proof of Smoothness for Morphic-Regularized Navier-Stokes Equations

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Step 1: Ensuring Non-Singularity

The first step toward proving smoothness is to ensure that the introduction of the morphic function prevents the formation of singularities. Singularities in fluid dynamics arise when the velocity field develops infinite gradients, leading to unbounded behavior. By introducing a morphic regularization term, we aim to modify the vortex core and other regions where singularities might form, smoothing out the velocity field and keeping the system well-behaved.

Smoothing with the Morphic Function

The morphic function introduces a regularization parameter ϵ that acts as a small buffer or smoothing term. This regularization prevents the velocity field from becoming infinite at critical points, such as the center of a vortex.

Consider the classical case of a vortex where the velocity field is given by:

$$u(r) = \frac{1}{r},$$

where r is the radial distance from the vortex center. As $r \rightarrow 0$, the velocity $u(r)$ diverges to infinity, creating a singularity at the vortex core.

To prevent this singularity, we modify the velocity field using the morphic regularization term ϵ , giving:

$$u_{\text{morphic}}(r) = \frac{1}{\sqrt{r^2 + \epsilon^2}},$$

where ϵ is a small positive constant that ensures the velocity remains finite as $r \rightarrow 0$.

Impact of the Regularization Term

The regularization term ϵ^2 acts as a buffer, preventing the velocity from becoming infinite at $r = 0$. Specifically:

$$u_{\text{morphic}}(0) = \frac{1}{\epsilon},$$

which remains finite as long as ϵ is non-zero. The presence of ϵ smooths out the velocity field near the singularity, ensuring that the velocity does not diverge.

For larger values of r , the regularized velocity field $u_{\text{morphic}}(r)$ behaves similarly to the classical vortex solution. As r increases, the influence of the ϵ^2 term diminishes, and the velocity field asymptotically approaches:

$$u_{\text{morphic}}(r) \approx \frac{1}{r} \quad \text{for } r \gg \epsilon.$$

Thus, the morphic regularization only significantly affects the region close to $r = 0$, where it prevents singular behavior.

Smoothing in High Reynolds Number Flows

In turbulent flows, where steep velocity gradients occur, the morphic regularization term $\epsilon \mathbf{R}(\mathbf{u})$ introduces additional dissipation, ensuring that the velocity gradients remain controlled. For high Reynolds number flows, the dissipation term is given by:

$$\frac{d}{dt}E(t) = -\nu \|\nabla \mathbf{u}\|_{L^2}^2 - \epsilon \|\nabla \mathbf{u}\|_{L^2}^2,$$

where $E(t)$ is the energy of the system. The second term, involving ϵ , increases the rate of dissipation in regions with steep gradients, preventing singularities from forming.

Conclusion of Step 1

By introducing the morphic function and the regularization parameter ϵ , we ensure that the velocity field remains finite, preventing the formation of singularities. This step lays the foundation for the overall proof of smoothness, addressing one of the key challenges in fluid dynamics: controlling the behavior of the system near singular points.